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Paleomagnetism of the Tertiary Clarno Formation of Central Oregon and Its Significance for the Tectonic History of the Pacific Northwest

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The Clarno Formation, a mostly Eocene and partly early Oligocene sequence of andesitic lavas and volcaniclastic rocks, is the oldest Tertiary formation exposed in north central Oregon. Remanent magnetization directions at 46 sites in the lavas provide a paleomagnetic pole at 84°N, 278°E with a 95% confidence cone of 7°. Comparison of this pole with the North American reference pole for Eocene time indicates that the Clarno Formation has rotated 16° clockwise with an uncertainty of 10° but has undergone no significant latitudinal displacement. Further comparison with paleomagnetic data from the Miocene Columbia River Basalt Group and Steens Basalt shows that large-scale block rotations in the eastern Columbia Embayment ceased sometime between 38 m.y. and 15 m.y. A paleogeographic reconstruction for 38 m.y. is offered which is consistent with the observed rotation of the Clarno and with the other paleomagnetic data from the Pacific Northwest, which are briefly reviewed. This reconstruction shows that it is possible to account for virtually all of the paleomagnetically indicated rotations in pre-Miocene Tertiary rocks of the Pacific Northwest by an extensional tectonic model. As a consequence, only part of the rotations in the pre-Tertiary rocks of the eastern Columbia Embayment need to be the result of accretionary tectonics. The initiation of magmatism in the eastern Columbia Embayment that is represented by the Clarno Formation itself may have been contemporaneous with the beginning of crustal extension in the Pacific Northwest.

INTRODUCTION

Recent paleomagnetic investigations have helped bring about a fundamental change in our understanding of the complicated tectonic history of the Pacific Northwest geographic province (abbreviated PNW, roughly the area in Figure 1). The pattern of structural trends and the distribution of pre-Tertiary rocks prompted early speculation by several authors [*Carey*, 1958; *Wise*, 1963; *Hamilton and Myers*, 1966] that large-scale rearrangements of extensive segments of the earth's crust had occurred in the PNW during Mesozoic and Tertiary time. Their basic observations were the comparatively greater width of the cordillera between 35° and 50° north latitude and the arcuate structural trends extending from within the Klamath Mountains northeastward through the pre-Tertiary rocks of the Blue Mountains uplift into western Idaho. They proposed that these features could be interpreted as distortion of structures that originally trended more nearly parallel to the rest of the cordillera, caused by some combination of northwest-directed right-lateral distributed shear and extensional tectonics. The predicted result was that rocks in the PNW should have anomalous paleomagnetic declinations, all rotated toward the east [*Hamilton and Myers*, 1966]. Paleomagnetic data supporting such rearrangements in the PNW and elsewhere were beginning to accumulate at the same time [*Cox*, 1957, *Beck*, 1962], but in general, the connections were missed. A notable exception was *Irving* [1964], who offered a very explicit interpretation of the data published by *Cox* [1957] for the Siletz River Volcanics in terms of oroclinal bending as is proposed by *Carey* [1958]. *Watkins* [1965*a*] also made an early attempt to interpret anomalous paleomagnetic declinations in sections of the Columbia River Basalt Group along the eastern margin of the Columbia Plateau as representing oroclinal bending of the sort envisaged by *Carey* [1958]. Watkins' conclusion was strenuously opposed by *Taubeneck* [1966], who argued that rotation was precluded by the uniform trend of the enormous Grande Ronde–Cornucopia (later called Chief Joseph) dike swarm, which is in the same area and is the same age as the basalt flows sampled by Watkins (see also *Watkins* [1967], *Taubeneck* [1967], and *Watkins and Baksi* [1974]). As it finally turned out, there is no convincing paleomagnetic evidence for oroclinal bending in the Columbia River Basalt Group [*Hooper et al.*, 1979], but as will be summarized later, older (pre-Miocene) rocks along the Blue Mountains axis show ample evidence for clockwise tectonic rotations, the amount increasing with the age of the rocks.

With the work of *Simpson and Cox* [1977], abundant paleomagnetic data from PNW rocks began to appear. A review of paleomagnetic data related to plate margin processes in the western cordillera has been given by *Beck* [1980]; here we present an updated summary of the data for most of the PNW. The paleomagnetically determined rotations are shown in Figure 1, and the supporting data, including paleolatitude anomalies, if present, are given in Table 1. These displacements have nearly all been recalculated from the original publications, and some new aspects of the recalculations will be discussed in a later section. Most (all but six) of the formations have been rotated by significant amounts, and the sense of all the rotations is clockwise. Only three of the rock units have notably anomalous paleomagnetic latitudes. These three are pre-Tertiary in age, have all come from farther south, all show large rotations, and are among the oldest rocks in their respective provinces; they are the Mount Stuart batholith (MS), the Seven Devils Group (SD), and the Huntington Formation (HV). Paleomagnetic data from Triassic rocks of the Seven Devils Group and the Huntington Formation, at the eastern edge of the Columbia Embayment, show that they are parts of the accreted allochthonous terrane known as Wrangellia [*Hillhouse et al.*, 1982]. Moreover, other paleomagnetic data show that part of the rotational history of these fragments of Wrangellia was shared by the Late Jurassic and Early Cretaceous batholiths that intrude them and other early Mesozoic rocks to the west [*Wilson and Cox*, 1980]. An implication of this result is that all of the pre-Cretaceous rocks of the Blue

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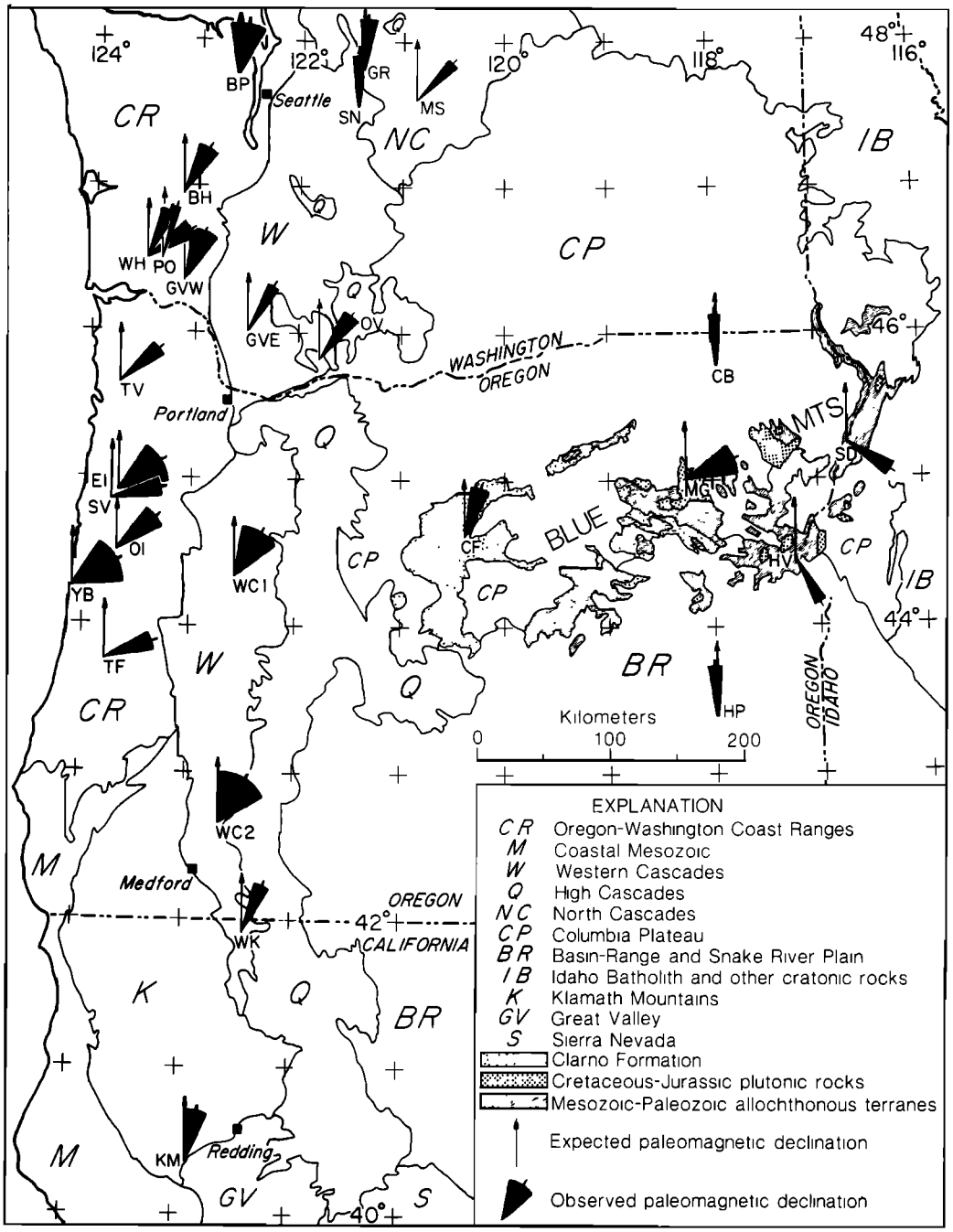


Fig. 1. Map of part of the Pacific Northwest showing geologic-physiographic provinces, the Eocene and Oligocene Clarno Formation, and pre-Tertiary rocks in the eastern part of the Columbia Embayment. Paleomagnetic declinations are shown with 95% confidence ranges; see Table 1 for identification of paleomagnetic data sites.

Mountains province of northeastern Oregon and westernmost Idaho are components of one or more accreted terranes, a possibility explicitly suggested by *Dickinson and Thayer* [1978] before any of the paleomagnetic data had become available.

The main intent of this article is to present new paleomagnetic data from the mostly Eocene and partly early Oligocene Clarno Formation in east central Oregon and to show how they bear on the tectonic history of the eastern province, which includes the North Cascades, the Columbia Plateau, the northwesternmost Basin Ranges, and the various components of the Blue Mountains uplift. We conclude that the Clarno Formation represents the last stage of large-scale coherent tectonic rotation in the eastern part of the Columbia Embayment. The paleomagnetic data indicate moderate clockwise rotation of the Clarno but no significant latitude shift. This rotation in turn allows the Clarno to be properly placed in a tectonic reconstruction of the PNW for the end of Eocene time. An important aspect of such a reconstruction is the necessity to accommodate the provenance of the sedimentary rocks of the Tyee basin in the southern and central Oregon Coast Range. It has recently been shown that after about 50 m.y. this provenance was most likely to have been the Idaho batholith and associated rocks [*Peterman et al.*, 1981; *Heller and Ryberg*, 1983; *Heller et al.*, 1985]. How close the Tyee basin could have been to its eastern source after completion of

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accretion of the Coast Range basement rocks and prior to the beginning of Western Cascades volcanic activity must be controlled to a great extent by the tectonic history of the pre-Miocene rocks in eastern and central Oregon, including the Clarno Formation. The new data from the Clarno Formation fill the gap in the paleomagnetic record from the eastern Columbia Embayment—a record of accretion (Seven Devils Group and Huntington Formation), rotation during the last stages of accretion (late Mesozoic granitic rocks), subsequent extension (Clarno Formation), and near stasis (Columbia River Basalt Group).

GEOLOGY

The Eocene and early Oligocene intermediate volcanic rocks of the Clarno Formation are exposed along the Blue Mountains uplift in central Oregon (Figures 1 and 2). The Clarno Formation consists predominantly of flows, breccias, and intrusions of andesite and basaltic andesite, along with lesser amounts of olivine basalt, dacite, and rhyolite. Interbedded with the flows are volcaniclastic tuffs, laharic breccias, and epiclastic siltstone, sandstone, and conglomerate [*Swanson*, 1969]. The measured thickness varies from place to place but approaches 2 km in the Horse Heaven mining district east of Ashwood [*Waters et al.*, 1951]. The Clarno Formation unconformably overlies Cretaceous marine sedimentary rocks and other older metasedimentary rocks [*Wilkinson and Oles*, 1968; *Oles and Enlows*, 1971] and in turn is unconformably overlain by the Oligocene and Miocene John Day Formation and Miocene Columbia River Basalt Group.

Eleven potassium-argon ages have been published for samples that are undisputably from the Clarno Formation; the ages are shown in Figure 2 as are summarized by *Fiebelkorn et al.* [1982]. The range is from 53.7 to 33.7 m.y., and the average age is 41.8 m.y. with a standard deviation of 6.6 m.y. In the Mitchell area the formation can be divided into flows of the lower part of the Clarno (46.1 to 37.5 m.y.) and flows of the upper part (about 34 m.y.), separated by an angular unconformity [*Enlows and Parker*, 1972]. In the northern outcrop area, ages for the upper part of the Clarno range from 42.1 m.y. in the Horse Heaven district [*Swanson and Robinson*, 1968] to 34.9 m.y. near Clarno [*Evernden and James*, 1964; *Evernden et al.*, 1964]. A single age of 53.7 m.y. was obtained in the southern part of the area north of Post [*Fiebelkorn et al.*, 1982]

The exposed part of the Clarno Formation may represent part of a continental magmatic episode that was active in the Eocene and early Oligocene. Chemically, the rocks form a calc-alkalic suite which is depleted in lithophilic elements when compared to other continental margin arcs [*Rogers and Ragland*, 1980]. Rogers and Ragland interpret the significance of this chemistry to be that the Clarno formed on a thin continental margin having a crustal thickness of 20 to 30 km. A detailed description of the central part of the Clarno was published by *Noblett* [1981], who showed on the basis of petrographic and chemical evidence that it was largely the product of subduction-related volcanism. *Armstrong* [1978] considered the Clarno to be part of the Challis volcanic episode (55 m.y. to 33 m.y.), during which activity was concentrated along an axis from northeastern Washington southeastward through Idaho to northwest Wyoming. However, andesitic volcanic rocks temporally equivalent to the upper part of the Clarno (42 m.y. and younger) crop out in south central Oregon and across central Nevada and Utah [*Walker*, 1963; *Wells*, 1979; *Stewart et al.*, 1977]. These volcanic rocks lie well to the west of the Challis axis and are in part correlative with the oldest volcanic rocks found in the Western Cascade arc [*Wells et al.*, 1984]. Thus the Clarno Formation seems to be transitional in time and space between Challis volcanism and the Western Cascade magmatic arc, and it may represent a remnant arc fragment isolated by interarc rifting.

PALEOMAGNETISM

Road cuts in the Clarno area are riddled with small drill holes, attesting to the attention of many paleomagnetists over the past few decades; the results of the earliest of these investigations (prior to 1962), however, have never come to light. The first report on paleomagnetism in the Clarno was by *Whitney* [1975], who drilled flows and intrusive rocks at several localities at and north of Mitchell (Figure 2). He concluded, from results for seven stable sites, that the Clarno rocks were magnetized nearly parallel to the late Eocene expected direction for North America and thus had not participated in any major postmagnetization tectonic rotations. We have combined several of Whitney's sites with our own data (see below). Whitney's work as well as early unpublished work by C. S. Gromme and new data by D. C. Engebretson and M. E. Beck, Jr., formed the basis of later calculations [*Beck et al.*, 1978] which also seemed to show the Clarno outcrop area to be not significantly rotated, though a somewhat easterly declination was reported. The results given in this paper are an outgrowth of the earlier work.

In all, we obtained paleomagnetic results from 46 sites in Clarno lava flows scattered over approximately 3500 km^2 (Figure 2). Many otherwise promising outcrops of lava were left unsampled because we were unable to obtain convincing measurements of structural attitude. The remanent magnetization data for each lava flow are given in Table 2.

Our 46 sites are the product of five separate sampling expeditions. Results from site 24 were obtained from oriented hand samples collected by C. S. Gromme in 1962. Fifteen hand samples were collected from six superposed lava flows. These flows include no intercalated sediments, share a common structural attitude, and yield almost identical remanence directions, so we have treated the mean of the 15 samples as a single measurement of the paleomagnetic field. Other data obtained in 1962 have been superseded and confirmed by the later work. Sites 18, 26, 36, and 44 are from *Whitney* [1975], who drilled from 7 to 13 samples per locality. We have combined samples from adjacent localities in the case of sites 18 and 26, because we felt after examining the outcrops that the same cooling units may have been sampled twice. The remaining 41 sites were sampled by the authors in 1977, 1978, and 1979; one of these (site 30) consists of combined results from three contiguous flows. Seven to nine 2.5-cm cores were drilled at each site. Azimuthal orientation was by magnetic compass using back-azimuths to a second compass to check for deflections arising from excessively strong magnetization of the outcrops. In all cases the deflections were small or nonexistent, indicating mainly that none of these outcrops had been extensively remagnetized by lightning.

All samples were magnetically cleaned by the alternating-field (AF) method. In general, the peak field strength used to clean each site was chosen after observing the response of two or more pilot specimens to progressive AF demagnetization at 10-mT intervals to a maximum of 100 mT. For the most part the Clarno lavas are magnetically stable, and the response of those that initially showed scattered directions was good.

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TABLE 1. Summary of Paleomagnetic

Map Symbol	Site, deg		Anomaly, deg		N	Pole, deg		σ_p
	Latitude North	Longitude East	ΔD	ΔL		Latitude North	Longitude East	
			Miocene					
PO	46.5	−123.4	14.7 ± 3.2	1.2 ± 2.7	1			
SN	47.6	−121.5	−0.1 ± 8.1	−5.1 ± 5.1	34	84.9	237.7	
CB	45.8	−118.0	−1.8 ± 6.3	−0.4 ± 4.4	59	88.7	171.6	16.9
HP	42.5	−119.2	−1.3 ± 8.1	−1.5 ± 5.8	29	88.3	209.0	18.2
			Oligocene					
OI	44.4	−123.7	47.6 ± 12.0	6.2 ± 9.3	14	60.9	325.6	22.1
GR	47.8	−121.4	5.0 ± 8.2	−2.3 ± 5.4	31	86.2	178.4	
OV	45.9	−122.0	35.7 ± 9.3	−3.0 ± 6.2	34	72.6	315.0	21.8
WC1	44.3	−122.6	30.1 ± 20.5	−14.9 ± 10.7	14	70.8	270.9	25.3
WC2	42.7	−122.7	29.6 ± 26.2	−8.0 ± 16.5	10	74.1	291.5	32.1
WK	41.9	−122.4	18.4 ± 7.8	2.9 ± 6.0	49	82.2	349.0	27.6
			Eocene					
BP	47.8	−122.7	7.0 ± 17.2	−1.7 ± 10.6	22	85.4	210.1	
BH	47.0	−123.2	26.5 ± 10.7	−0.6 ± 6.9	35	78.8	305.0	25.1
WH	46.5	−123.5	20.3 ± 7.8	6.1 ± 5.6	39	81.9	349.2	21.6
WH	46.5	−123.5	52.4 ± 10.2	6.8 ± 7.4	12	59.8	319.0	16.5
GVW	46.4	−123.1	22.5 ± 14.4	9.9 ± 11.3	10	77.2	5.7	22.5
YB	44.3	−124.1	49.0 ± 22.5	1.5 ± 16.0	9	62.0	312.4	29.7
EI	44.9	−123.7	53.0 ± 23.9	1.3 ± 16.3	10	60.1	308.4	31.9
TV	45.6	−123.7	45.3 ± 8.7	2.2 ± 6.0	32	65.5	312.6	21.6
TF	43.8	−123.8	67.4 ± 14.5	−3.6 ± 9.3	40	50.9	296.1	
SV	44.9	−123.8	69.5 ± 9.1	4.2 ± 6.6	33	48.0	306.5	24.4
GVE	46.0	−122.5	28.0 ± 7.0	7.0 ± 5.1	37	75.5	345.5	18.4
CF	44.6	−120.4	15.7 ± 9.5	−2.6 ± 6.3	46	83.9	277.6	27.6
			Cretaceous					
MS	47.6	−120.9	43.2 ± 6.2	29.4 ± 5.0	17	68.3	35.3	
KM	40.4	−122.9	11.5 ± 12.3	7.7 ± 12.4	21			
			Jurassic and Cretaceous					
MG	45.1	−117.9	66.4 ± 16.5	4.2 ± 11.4	9	68.5	320.9	
			Triassic					
SD	45.2	−116.7	124.2 ± 8.6	45.3 ± 7.9	46	29.9	155.6	
HV	44.5	−117.2	153.7 ± 6.3	39.0 ± 5.9	26	46.5	125.9	

ΔD is declination anomaly, positive eastward; ΔL is apparent latitude shift, positive northward; uncertainty is at 95% confidence as is modified by *Demarest* [1983]. N is number of lava flows, sampling sites, or specimens (see text). The value σ_p is angular standard deviation of VGPs. Age is radiometrically determined; N.D., not determined. Reference pole: M, the mean for the Pomona flow on the Columbia Plateau; G, the geographic north pole with a 95% confidence of 4 degrees; O and E, Oligocene and Eocene reference poles from *Diehl et al.* [1983]; K, Cretaceous reference pole from *Mankinen* [1978].

*As is recalculated by *Hillhouse et al.* [1982].

†See *Hillhouse et al.* [1982].

‡Recalculated giving specimens' unit weight.

Figure 3 shows the reduction in angular standard deviation produced by the AF cleaning. Peak fields used for the AF cleaning ranged from 15 to 80 mT, but the majority of sites were cleaned in fields of 20 to 40 mT.

Mean remanence directions were compensated for postmagnetization tilt of up to 37° by rotating about the strike of bedding so that the dip of bedding became zero. The corrected directions are shown in Figure 4. Although most of these structural corrections were small (more than half were less than 10°; see Table 2), they are probably the largest single source of error in determining the mean magnetization direction of the Clarno Formation. Wherever possible the structural attitudes were measured on nearby sedimentary rocks, and these corrections are accurate to within a few degrees. In many cases, however, it was necessary to use the evidence of attitude from the lava flows themselves, for example, the dips of flow tops or between-flow boundaries, pervasive platy flow structure, or rarely, the attitudes of columnar joints. Some of these corrections may be wrong by as much as 10°. The problem of initial dip also complicates the search for the proper structural correction, because some of the dips observed in the field may represent solidification of the lava flow on an inclined surface rather than postmagnetization tectonic tilt. Overall mean remanence directions for the Clarno Formation are given in Table 3, both before and after structural correction. The between-site dispersion decreases slightly after structural correction, and the overall mean direction does not change significantly. That the result of the "fold test" is not more strongly positive may be partly attributed to the problems mentioned above but also to the fact that the Clarno is not much deformed; the angular standard deviation of poles to bedding is only 15°.

Both magnetic polarities are found in the Clarno Formation. Figure 4 and Table 3 show that the mean directions of 15 normal and 31 reversed sites are virtually antiparallel. This implies that there is no uniform secondary magnetization remaining in the specimens after AF cleaning. The magnetic polarity of each site is shown in Figure 2. We can discern no simple map distribution of normal and reversed sites; this observation coupled with the long duration of Clarno volcanic activity implied by the potassium-argon ages leads us to believe that more than a few different polarity intervals have been sampled.

The difficulties in making accurate structural corrections

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Data for Part of Pacific Northwest

Rock Unit	Age, m.y.	Reference Pole	References	
			Pole	Age
Pomona flow	12	M	*Magill et al.* [1982]	*McKee et al.* [1977]
Snoqualmie batholith	17–18	G	*Beske et al.* [1973]	*Yeats and Engels* [1971]
Columbia River Basalt Group	14–17	G	*Choinere and Swanson* [1979], *Hooper et al.* [1979], *Rietman* [1966], *Watkins and Baksi* [1974]	*McKee et al.* [1977], *Watkins and Baksi* [1974], *McKee et al.* [1981]
High Plateaus lavas	15.5	G	*Watkins* [1965*b*], E. A. Mankinen et al. (written communication, 1983)	*Baksi et al.* [1967]
Basaltic sills	30–32	O	*Beck and Plumley* [1980], *Plumley* [1980]	*Fiebelkorn et al.* [1982]
Grotto batholith	25–26	O	*Beske et al.* [1973]	*Yeats and Engels* [1971]
Ohanapecosh Volcanics	31–37	O	*Bates et al.* [1981], *Bates* [1980]	*Bates et al.* [1981], *Bates* [1980]
Western Cascades Volcanics	20–30	O	*Magill and Cox* [1980]	*Magill and Cox* [1980]
Western Cascades Volcanics	20–29	O	*Magill and Cox* [1980]	*Fiebelkorn et al.* [1982]
Western Cascades Volcanics	23–33	O	*Craig et al.* [1981], *Beck et al.* [1986]	*Craig et al.* [1981], *Beck et al.* [1986]
Crescent Formation	55	E	*Beck and Engebretson* [1982]	*Duncan* [1982]
Crescent Formation	52–56	E	*Globerman et al.* [1982]	*Duncan* [1982]
Crescent Formation	48–52	E	*Wells and Coe* [1985]	*Duncan* [1982]
Crescent Formation	48–52	E	*Wells and Coe* [1985]	*Duncan* [1982]
Goble Volcanics	39	O/E	*Wells and Coe* [1985]	*Wells and Coe* [1985]
Yachats Basalt	37	O/E	*Simpson and Cox* [1977], *Simpson* [1977]	*Fiebelkorn et al.* [1982]
Basaltic sills	N.D.	E	*Beck and Plumley* [1980], *Plumley* [1980]	*Beck and Plumley* [1980], *Plumley* [1980]
Tillamook Volcanics	43–46	E	*Magill et al.* [1981]	*Magill et al.* [1981]
Tyee Formation	N.D.	E	*Simpson and Cox* [1977], *Simpson* [1977]	*Simpson and Cox* [1977]
Siletz River Volcanics	53–56	E	*Simpson and Cox* [1977], *Simpson* [1977]	*Duncan* [1982]
Goble Volcanics	32–45	O/E	*Beck and Burr* [1979]	*Beck and Burr* [1979]
Clarno Formation	34–54	E	this study	
Mt. Stuart batholith	83–93	K	*Beck et al.* [1981]	*Engels and Crowder* [1971]
Great Valley sequence	N.D.	K	*Mankinen and Irwin* [1982], *Irwin et al.* [1984]	*Mankinen and Irwin* [1982]
Granitic intrusive rocks	130–150	†	*Wilson and Cox* [1980]*	*Wilson and Cox* [1980]*
Seven Devils Group (part)	N.D.	†	*Hillhouse et al.* [1982]‡	*Hillhouse et al.* [1982]‡
Huntington Formation	N.D.	†	*Hillhouse et al.* [1982]‡	*Hillhouse et al.* [1982]‡

mentioned above are characteristic of many parts of the PNW, owing to limited exposure and lack of sedimentary interbeds [*Beck and Burr*, 1979; *Globerman et al.*, 1982]. In volcanic formations of intermediate and felsic composition this problem is made worse by the high viscosity of the lava, which commonly causes the congealed flows to have irregular and nonplanar shapes. In the following section we will attempt to evaluate the uncertainties in structural corrections by comparing the between-site angular dispersion of the Clarno data with other paleomagnetic data.

The observed angular standard deviation of VGPs for the Clarno Formation after structural correction is 27.6°. The within-site dispersion may be considered as the contribution by nonsystematic orientation errors and limitations of magnetometer precision and by the intrinsic inaccuracy of lava flows as magnetic recorders; its average value is 11.0° when converted to VGP space. Removing this contribution leaves 25.3° as the value of dispersion caused by all other sources including geomagnetic secular variation; the 95% confidence limits are at 21.2° and 31.2° [*Cox*, 1969]. A dispersion of 25.3° is significantly greater than the amplitude of secular variation for the past 5 m.y., as is derived from worldwide paleomagnetic data and summarized by *McFadden and McElhinny* [1984]. In their Table 5, for the paleomagnetic latitude of the Clarno Formation we find a value of 16.7° with lower and upper 95% confidence limits at 15.6° and 18.0°, respectively. If we assume that the geomagnetic secular variation had the same amplitude when the Clarno lavas were erupted as it had during the past 5 m.y., then the contribution to the angular standard deviation of Clarno VGPs that is not attributable to secular variation and within-site dispersion is 19.0°, which becomes 13.4° when transformed back to direction space [*Cox*, 1970]. This value seems higher than would result only from the structural uncertainties mentioned above. A second source of uncertainty in the structural correction might be caused by the existence of steeply plunging fold axes rather than the horizontal fold axes that we have assumed at all sites, but geologic mapping shows only subhorizontal fold axes (Figure 2). The question raised by this discrepancy is whether greater dispersion of VGPs is to be expected in rock units that have been subjected to large-scale tectonic rotations.

From Table 1 we see that the range of observed angular standard deviations of VGPs for Eocene and Oligocene volcanic rocks in the PNW extends from 16.5° to 32.1°, while the two corresponding values for Miocene flood basalts are 16.9° and 18.2°. Another comparison can be made using the reference poles. The average value of angular standard deviation of VGPs in each of the eight Paleocene and Eocene study areas of *Diehl et al.* [1983] is 16.6° with a range from 12.8° to 20.7°, similar to the Miocene and 5-m.y. values mentioned above. The comparison is appropriate because the paleomagnetic latitudes of the PNW data and the Montana volcanic province studied by Diehl et al. are nearly the same. For the overall reference poles calculated by *Diehl et al.* [1983], the angular standard deviations of VGPs are somewhat larger in two of three cases; 22.7° for the Oligocene pole ($N = 102$) and 20.2°

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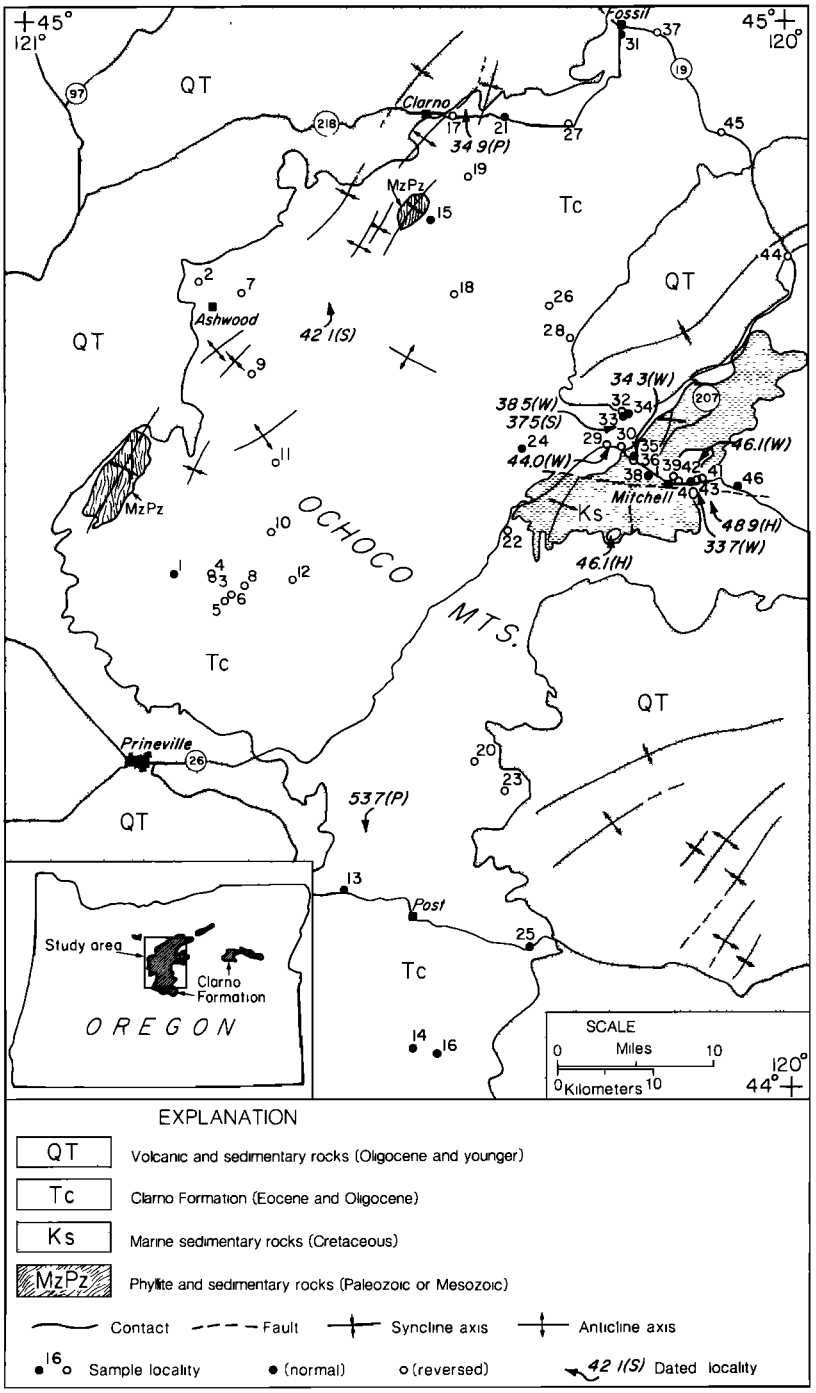


Fig. 2. Simplified geologic map showing paleomagnetic sample sites in the Clarno Formation. Locations of samples for which ages have been published are also shown with ages in millions of years and materials dated (W, whole rock; S, sanidine; P, plagioclase; and H, hornblende). Geology from *Swanson* [1969].

for the Eocene pole ($N = 147$). Thus the dispersion of VGPs observed in the Clarno Formation and other pre-Miocene rocks of the PNW is somewhat greater than would be expected from geomagnetic secular variation alone. This difference must be due largely to imperfect restoration of each site to its predeformation attitude, although unremoved components of secondary magnetization are always potential contributors.

DISCUSSION

The paleomagnetic pole calculated from structurally corrected data for the 46 sites in the Clarno Formation is shown in Figure 5. Also shown in Figure 5 are poles for the North American craton for Cretaceous and Tertiary time. The Clarno pole differs significantly from the Eocene reference pole, and the difference results almost entirely from a more easterly declination for the Clarno than would be predicted from the reference pole. In an earlier abstract [*Beck et al.*, 1978] we published a mean paleomagnetic direction for 13 sites in the Clarno Formation. The declination we reported was 9.5°, which is 4° more easterly than the value given in Table 3, but the 95% confidence cone of the earlier result was 65% larger than the value in Table 3. We had concluded at that time that the Clarno was not significantly rotated in

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TABLE 2. Site-Mean Paleomagnetic Data From the Clarno Formation

Site	Strike, deg	Dip, deg	$N(N_c)$	H, mT	I, deg	D, deg	R	k	α_{95}, deg	λ, deg	ϕ, deg
1	076	22	9(9)	20	62.4	015.3	8.8510	53.7	7.1	79.0	327.8
2	130	22	8(8)	20	−74.6	087.8	7.9778	315.8	3.1	−37.1	022.0
3	105	08	8(8)	40	−71.8	189.2	7.9381	113.0	5.1	−76.5	081.4
4	105	08	8(8)	80	−63.1	214.4	7.9854	480.1	2.5	−65.7	136.8
5		00	8(8)	80	−35.7	162.9	7.5270	14.8	14.9	−61.5	274.7
6	120	06	8(8)	60	−45.0	184.1	7.9648	198.9	3.9	−71.8	227.4
7		00	8(8)	40	−65.9	160.3	7.8757	56.3	7.5	−76.1	350.5
8	130	14	6(8)	60	−29.2	175.2	5.9104	55.8	9.1	−60.9	248.8
9	220	18	8(8)	40	−80.8	152.5	7.9272	96.1	5.7	−59.7	042.9
10	032	08	7(8)	80	−47.6	183.9	6.9580	142.8	5.1	−73.9	226.9
11	045	21	8(8)	20	−63.0	237.5	7.9928	972.2	1.8	−49.9	128.5
12		00	8(8)	60	−42.2	176.8	7.9706	237.7	3.6	−69.7	247.8
13	220	05	8(8)	20	52.5	022.7	7.9547	154.5	4.5	69.2	353.9
14	321	10	7(8)	40	68.1	023.2	6.9145	70.2	7.3	72.9	296.6
15	335	26	8(8)	30	59.5	334.3	7.9736	264.9	3.4	70.6	145.2
16	310	06	4(8)	20	54.4	034.7	3.9755	122.4	8.3	62.0	336.5
17		00	7(8)	40	−56.7	157.7	6.9542	131.0	5.3	−71.6	312.9
18*	300	12	24(24)	20	−75.6	172.1	23.9505	464.7	1.4	−71.4	048.2
19		00	8(8)	40	−70.8	195.8	7.9716	246.7	3.5	−75.6	098.4
20	105	20	6(8)	40	−59.3	268.5	5.9246	66.3	8.3	−27.7	119.3
21		00	7(7)	30	62.5	034.5	6.9521	125.2	5.4	65.5	319.8
22	010	05	7(7)	20	−81.5	221.3	6.9728	220.7	4.1	−55.6	079.2
23	092	26	8(8)	20	−77.9	010.7	7.8597	49.9	7.9	−21.4	055.1
24*	282	18	52(52)	20	52.7	335.1	50.4036	32.0	3.6	67.7	127.7
25	120	09	8(8)	15	59.1	008.9	7.8896	63.4	7.0	82.1	359.4
26*	047	20	26(26)	20	−70.3	219.0	25.6270	67.0	3.5	−63.3	114.5
27	215	24	8(8)	30	−58.5	170.2	7.9407	118.1	5.1	−80.8	295.0
28	045	20	8(8)	20	−71.4	197.0	7.9208	250.8	3.5	−74.4	097.1
29	275	18	7(7)	30	−09.4	181.0	6.8833	51.4	8.5	−50.1	238.2
30*	232	37	31(34)	40	−56.6	196.7	29.7082	23.2	5.5	−75.4	174.5
31	240	20	8(8)	30	73.5	269.9	7.9668	210.9	3.8	37.4	199.9
32	315	15	8(8)	20	−50.6	219.1	7.6316	19.0	13.0	−56.8	159.9
33	245	20	7(7)	20	62.7	008 8	6.9841	378.1	3.1	83.7	331.5
34	240	35	7(7)	30	72.5	359.2	6.9509	122.8	5.5	76.9	237.9
35		00	7(7)	30	60.1	037.8	6.8915	55.3	8.2	62.3	323.8
36		00	8(8)	20	−81.3	229.1	7.9215	114.6	5.9	−54.0	082.1
37	320	09	8(8)	20	−47.6	192.2	7.9663	207.7	3.9	−71.1	205.0
38		00	6(7)	30	69.9	011.1	5.9628	134.6	5.8	78.3	273.9
39	045	05	4(4)	40	−69.5	118.4	3.8909	27.5	17.8	−49.9	005.0
40	270	05	6(6)	30	62.1	344.3	5.9747	197.6	4.9	78.7	149.3
41	000	05	7(7)	40	−70.0	191.7	6.8962	57.8	8.0	−77.9	094.7
42	045	05	7(7)	30	−68.0	163.0	6.9737	228.1	4.0	−76.9	005.7
43	180	05	7(7)	40	−74.7	172.7	6.9742	232.1	4.0	−72.7	048.0
44		00	9(9)	20	−62.9	209.2	8.8499	66.6	7.1	−69.3	140.4
45		00	7(7)	40	−26.0	172.0	6.9920	994.7	2.2	−58.1	254.7
46	210	17	6(7)	40	62.7	348.8	5.9936	776.4	2.4	82.0	150.5

Sites are locations shown in Figure 2. Strike and dip are attitude of original horizontal at outcrop; azimuth of dip is 90° clockwise from strike. $N(N_c)$, number of specimens used (collected). H, peak alternating field used for demagnetization. I and D, inclination (positive downward) and declination (eastward) of mean north-seeking magnetization at site after demagnetization and correction for dip and strike. R, resultant of N unit vectors. Value k, Fisher precision parameter. Value α_{95}, semiangle of 95% confidence cone. Values λ and ϕ, latitude (positive north) and east longitude of virtual geomagnetic pole.

*Average of several sites or superposed lava flows.

paleomagnetic terms. The rotation calculated from the new data is 15.7° ± 9.5° clockwise, which is significant at the 95% confidence level. The difference is partly due to the great increase in the number of sites and is also partly due to our use of the new Eocene reference pole for the North American craton recently published by *Diehl et al.* [1983]. A brief discussion of the methodology of calculation of tectonic rotations and translations and the associated confidence limits is in order at this point.

There are two basic ways of doing the calculation. The first is in "direction space", comparing observed and predicted declinations and inclinations, where the predicted values are calculated from a reference pole derived from rocks in stable continental areas. The second method is in "VGP space," comparing observed paleomagnetic poles with the stable reference poles. The choice of method depends on the nature of the problem to be solved. If the comparison is made in direction space, one obtains a declination anomaly directly interpretable as a tectonic rotation and an inclination anomaly or "flattening" [*Beck*, 1980]. If the comparison is made in VGP space (see Figure 5), one obtains a declination anomaly or rotation (the angle at the site subtended by the two poles) exactly as before and a difference in the distances from the site to the two poles which is equivalent to a paleomagnetic latitude anomaly. Inclination anomalies may have several causes, ranging from tectonic transport to initial dips in lava flows erupted on the flanks of volcanos or the effects of magnetic anisotropy in the rocks. When considering plate tectonic matters such as the transport of allochthonous tectonostratigraphic terranes, the common procedure is to present

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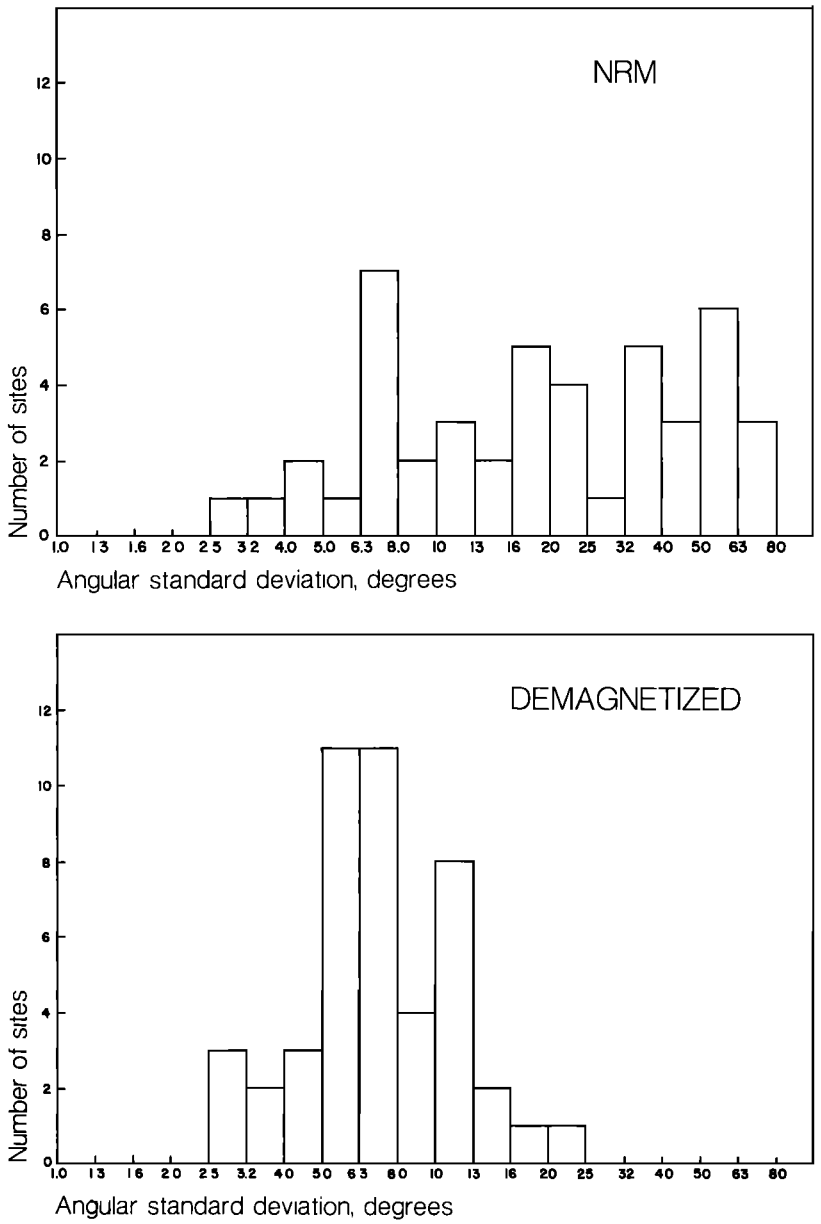


Fig. 3. Histograms showing reduction in angular standard deviation of magnetization directions at all sites as a result of alternating-field demagnetization. NRM is natural remanent magnetization as was first measured.

the data in the form of paleolatitude anomalies. Confidence limits are customarily taken as symmetrical about the mean value, whether one uses inclination or paleomagnetic latitude, and the transformation is made using the first derivative of the dipole formula. Because of the nonlinearity of the dipole formula, however, this method of calculation can result in a marginally significant inclination anomaly becoming insignificant when transformed into a paleolatitude anomaly if the uncertainty is more than a few degrees.

For the sake of uniformity we have recalculated the PNW rotations and displacements listed in Table 1 in terms of VGPs when this was not done in the original publications. Where necessary, we have referred to unpublished dissertations. In most cases the recomputation was straightforward: VGPs were calculated from site-mean directions and site coordinates, and the uncertainties in rotation and latitudinal displacement were calculated assuming that the VGPs for each study area (and the reference poles) are circularly distributed. For the Goble Volcanics (GVW and GVE) and Yachats Basalt (YB) we used an average of the Eocene and Oligocene reference poles of *Diehl et al.* [1983] to match the radiometric ages of the rocks. Following *Demarest* [1983], we have multiplied the combined uncertainties by 0.78 to convert bivariate confidence limits to univariate limits. We show a combined result for the Eocene lavas at Bremerton and Port Townsend (BP) in Washington; the large uncertainty reflects the fact that a large difference in paleomagnetic declination exists between the two areas. This difference is evidently a result of the formation of the Olympic "orocline" [*Beck and Engebretson*, 1982] but is too local to show at the scale of Figure 1. The data from the Western Cascades in Oregon (WC1 and WC2) were originally presented as a combined result by *Magill and Cox* [1980], but we have divided them because the two areas are separated by 1.7° of latitude.

In contrast to some previous reviews [*Wilson and Cox*, 1980; *Magill and Cox*, 1980], we find no indication of large-

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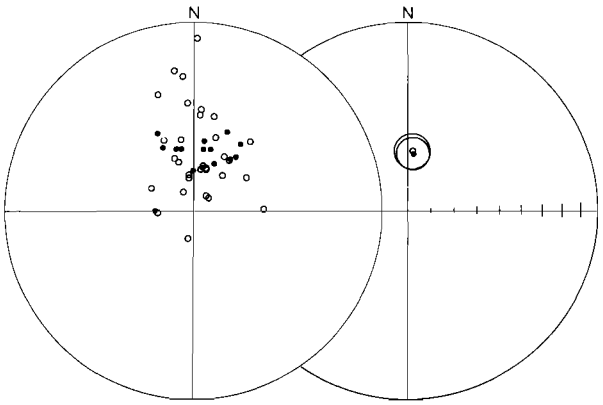


Fig. 4. Equal-area projections on lower hemisphere of magnetization directions after AF cleaning: (*left*) mean directions at individual sites (Table 2); (*right*) means of all normal and all reversed site-means with 95% confidence circles (Table 3). Solid circles are north-seeking directions; open circles are south-seeking directions.

scale post-Miocene rotation in the eastern Columbia Embayment or the northwesternmost Basin and Range Province. With the exception of the Picture Gorge Basalt and the section at Owyhee Ridge, essentially all of the paleomagnetic data published by *Watkins and Baksi* [1974], which were used extensively in these reviews, have been superseded by later work. We have calculated a new paleomagnetic pole for the Columbia River Basalt Group using data published by *Hooper et al.* [1979] and *Choinere and Swanson* [1979] and have included only the Picture Gorge data of *Watkins and Baksi* [1974]. The sampling area extends from 116.5° to 120° west longitude and from 44.5° to 46.7° north latitude, mostly north of but partly within the Blue Mountains uplift. We have included all the formations within the group except for the Saddle Mountains Basalt, which is significantly younger and volumetrically least important, and have taken precautions not to duplicate any data. For a reference pole we have used the geographic pole (see below) with an arbitrary uncertainty of 4°. A more detailed accounting of this calculation will be given elsewhere (E. A. Mankinen, et al., manuscript in preparation, 1986). From the results in Table 1 (CB) we conclude that there is no paleomagnetic evidence for regional post-Miocene tectonic rotation in the eastern part of the Columbia Plateau at the 95% confidence level. This is not to say that there is no paleomagnetic evidence for post-Miocene deformation in the Columbia River Basalt Group. *Sheriff* [1984]

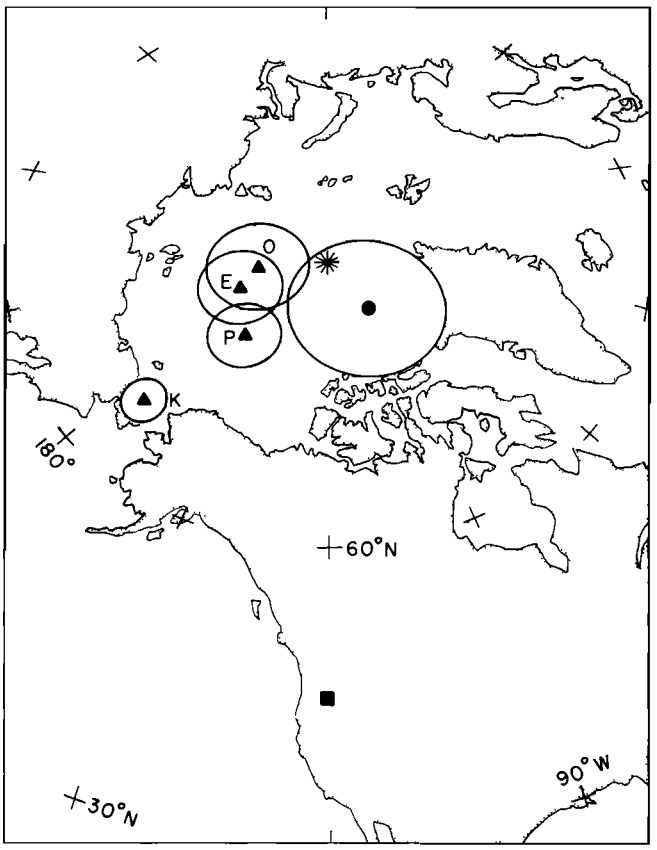


Fig. 5. Equal-area map of part of northern hemisphere showing paleomagnetic poles. Center of projection is outcrop area of Clarno Formation, shown by a square. Circle is paleomagnetic pole for Clarno Formation (46 sites, corrected for structure; see Table 3). Triangles are reference poles for North American craton (K, Cretaceous; P, Paleocene; E, Eocene; and O, Oligocene). Cretaceous pole is from *Mankinen* [1978]; the other three are from *Diehl et al.* [1983].

has made a systematic study of the Ginkgo flows of the Frenchman Springs Member of the Wanapum Basalt, the age of which is about 14 m.y. Sheriff presents evidence for progressively increasing clockwise rotation along a transect northwest of the Blue Mountains axis, starting from about 120° west longitude and 47° north latitude and proceeding southwest to the coast at Cape Foulweather, Oregon. The total differential rotation is of the order of 35°, and the differential rotation within the western Columbia Plateau is about 20°. On the

TABLE 3. Mean Magnetization Directions and Paleomagnetic Poles for the Clarno Formation

	N	I, deg	D, deg	r	k	α_{95}, deg	λ, deg	ϕ, deg	R	K	A_{95}, deg
					*All Sites**						
Uncorrected	46	65.0	6.9	42.6921	13.6	5.9	82.2	279.7	39.9993	7.5	8.2
Corrected	46	64.4	5.5	43.1367	15.7	5.5	83.9	277.6	41.0011	9.0	7.4
					Normal Polarity Sites						
Uncorrected	15	70.5	11.7	13.9624	13.5	10.8	75.3	268.7	12.5981	5.8	17.4
Corrected	15	65.1	6.0	14.5614	31.9	6.9	84.9	286.6	13.9884	13.8	10.7
					Reversed Polarity Sites†						
Uncorrected	31	62.3	5.2	28.8356	13.9	7.2	85.0	294.5	27.5453	8.7	9.3
Corrected	31	64.1	5.2	28.5736	12.4	7.7	83.3	274.1	27.0184	7.5	10.1

N, number of sites. I and D, mean inclination and declination. Values r and R, resultants of N unit vectors for directions and VGPs, respectively. Values k and K, Fisher precision parameters for directions and VGPs, respectively. Values α_{95} and A_{95}, semiangles of 95% confidence cones around mean directions and VGPs, respectively. Values λ and ϕ, north latitude and east longitude of paleomagnetic pole.

*Mean site location is 44.62°N, 239.61°E.

†Data are for south-seeking magnetization directions.

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other hand, a similar study by *Magill et al.* [1982] of the somewhat younger (12 m.y.) Pomona Member of the Saddle Mountains Basalt showed no evidence for differential rotation within the Columbian Plateau, but did show a significant rotation in the correlative basalt of Pack Sack Lookout west of the Cascades axis (locality PO in Figure 1). On a more detailed scale, *Reidel et al.* [1984] have used the Pomona Member of the Saddle Mountains Basalt to show that clockwise rotations are to be found on anticlinal ridges relative to the paleomagnetic declinations found in synclines. The relative rotations are usually of the order of 10° but may reach 30°. The area investigated by Reidel et al. is the Pasco Basin, within the Yakima Fold Belt province of the west central Columbia Plateau. These investigations serve to show that the western Columbia Plateau cannot be considered as a strictly rigid structural block for post-Miocene time. Nonetheless, the eastern part of the Plateau evidently may be so considered, at least within the numerical uncertainties typical of paleomagnetic data.

A large amount of new data has also been recently obtained from the Steens Basalt of the High Plateaus of Oregon [*Prevot et al.*, 1983; *Mankinen et al.*, 1985]. We have used this collection of directions (obtained from the Steens Mountain escarpment and the vicinity of Adel) together with those from one section at Abert Rim published by *Watkins* [1965*b*] to compute a Miocene paleomagnetic pole that is broadly representative of the northwesternmost Basin and Range Province. In so doing, we have explicitly avoided duplication of data and have also excluded all data from the transitional interval originally described in detail by *Watkins* [1969]. The data set will be fully described elsewhere (E. A. Mankinen, et al., manuscript in preparation, 1986). The result in terms of rotation assuming a reference pole at geographic north (Table 1, HP) is virtually identical to that from the Columbia River Basalt Group; that is, no significant rotation is observed. We have omitted the data from the section of lavas at Owyhee Ridge in easternmost Oregon from our calculation; there the mean declination in 14 superposed lavas is 25° [*Watkins and Baksi*, 1974]. This may well represent a real tectonic rotation of more or less local extent and warrants further investigation.

Use of the geographic pole as a reference for middle Miocene time is an assumption on our part and may seem unsupported; from Figure 5 one might predict a 15-m.y. pole midway between the Oligocene reference pole and the geographic pole. When the data from the Columbia River Basalt Group and the Oregon High Plateaus are subtracted from the total available, not very much is left for that time. We have made a simple average of four middle Miocene paleomagnetic poles for the North American craton; these are poles 14, 15, 16, and 18 cited in Table 1 of *Irving and Irving* [1982] and range in age from 12 to 17 m.y. The average pole is at 88.7° north, 141.1° east, with a 95% confidence radius of 7.2° for $N = 4$. This pole differs from the Columbia River Basalt Group pole by 0.7° and from the High Plateaus pole by 1.7°. We conclude that there is no discernible difference between the Columbia River Basalt Group pole, the High Plateaus pole, the reference pole computed from middle Miocene data from the North American craton, and the geographic pole.

TECTONIC INTERPRETATION

Many maps have been published showing tectonic reconstructions of the PNW for various ages; our version for the end of Eocene time (38 m.y., which is also close to the culmination of Clarno volcanism) is shown in cartoon form as Figure 6. The principles underlying this reconstruction are to satisfy insofar as possible the paleomagnetically determined rotations of rocks younger than the volcanic basement of the Coast Ranges and to satisfy sedimentary and volcanic overlap relationships. These relationships require the Oregon and Washington Coast Ranges to have been adjacent to the proto-Western Cascade arc and the surrounding pre-Tertiary rocks of the Klamath Mountains and North Cascade Range during Clarno time [*Heller and Ryberg*, 1983; *Wells et al.*, 1984]. In general, the clockwise rotations of rocks in the Blue Mountains province (Figures 1 and 6) occurred around an eastern pivot point constrained by geologic relationships along the Idaho border.

In this reconstruction, all of the rotations recorded in rocks younger than the Coast Ranges basement (Crescent Formation and Siletz River Volcanics) are the result of crustal extension; only the small remaining rotation of the basement rocks could have resulted from earlier collision or accretion. The main evidence supporting this conclusion comes from the Tyee basin, the Tertiary sedimentary basin that overlies the Eocene volcanic basement of the southern and central Oregon Coast Range. Sedimentologic and geochemical investigations of the sequence of formations in the basin demonstrate a profound change in provenance at about 50 m.y. The older and more deformed Roseburg Formation had a Klamath Mountains provenance, but during the interval from 50 to 45 m.y. (during deposition of the Bateman Formation and Tyee Formation, including its Elkton Siltstone Member) the provenance could not have been the Klamath Mountains province but was most likely the area of the Idaho batholith to the east [*Peterman et al.*, 1981; *Heller and Ryberg*, 1983; *Heller et al.*, 1985]. Because the sedimentary rocks of the Tyee basin have rotated approximately the same amount as the underlying seamount terrane which comprises the basement of the Coast Ranges, the implication is that all or nearly all of the rotation occurred after accretion of the basement was completed [*Heller and Ryberg*, 1983]. If so, the rotation represents westward movement of the southern end of the basin away from the source terrane; consequently, in our reconstruction we keep the southern end of the Coast Range as far east as possible. We also keep the north end of the Coast Ranges as far east as possible to accommodate the presence of continentally derived sedimentary rocks overlying Eocene volcanic rocks of the Olympic Peninsula and interbedded in the Tillamook Volcanics and in the volcanic rocks of the Western Cascades in Washington [*Hammond*, 1979]. Loosely linking the two parts of the Coast Ranges and the Klamath Mountains takes account of the continuity of the volcanic basement of the Coast Ranges and of the sedimentary overlap of the Tyee basin on the coastal Mesozoic province west of the Klamath Mountains. Because there is no paleomagnetic latitude anomaly recorded in the Clarno Formation, the Clarno must be left in place between the Coast Ranges block and the pre-Tertiary rocks of the Blue Mountains province, rather than shifting it to a more northerly place of origin, for example, which would have allowed a somewhat closer connection between the Tyee basin and the Idaho batholith.

The east side of the reconstruction is constrained by the geologic relationships in western Idaho described by *Hamilton* [1963]. Between latitudes 45.1° and 45.5° the Seven Devils Group is bounded on the east by a thrust fault, the Rapid River thrust. An assemblage of metamorphic rocks, the Riggins Group, forms the hanging wall of the thrust on the east and is intruded by Cretaceous rocks of the Idaho batholith.

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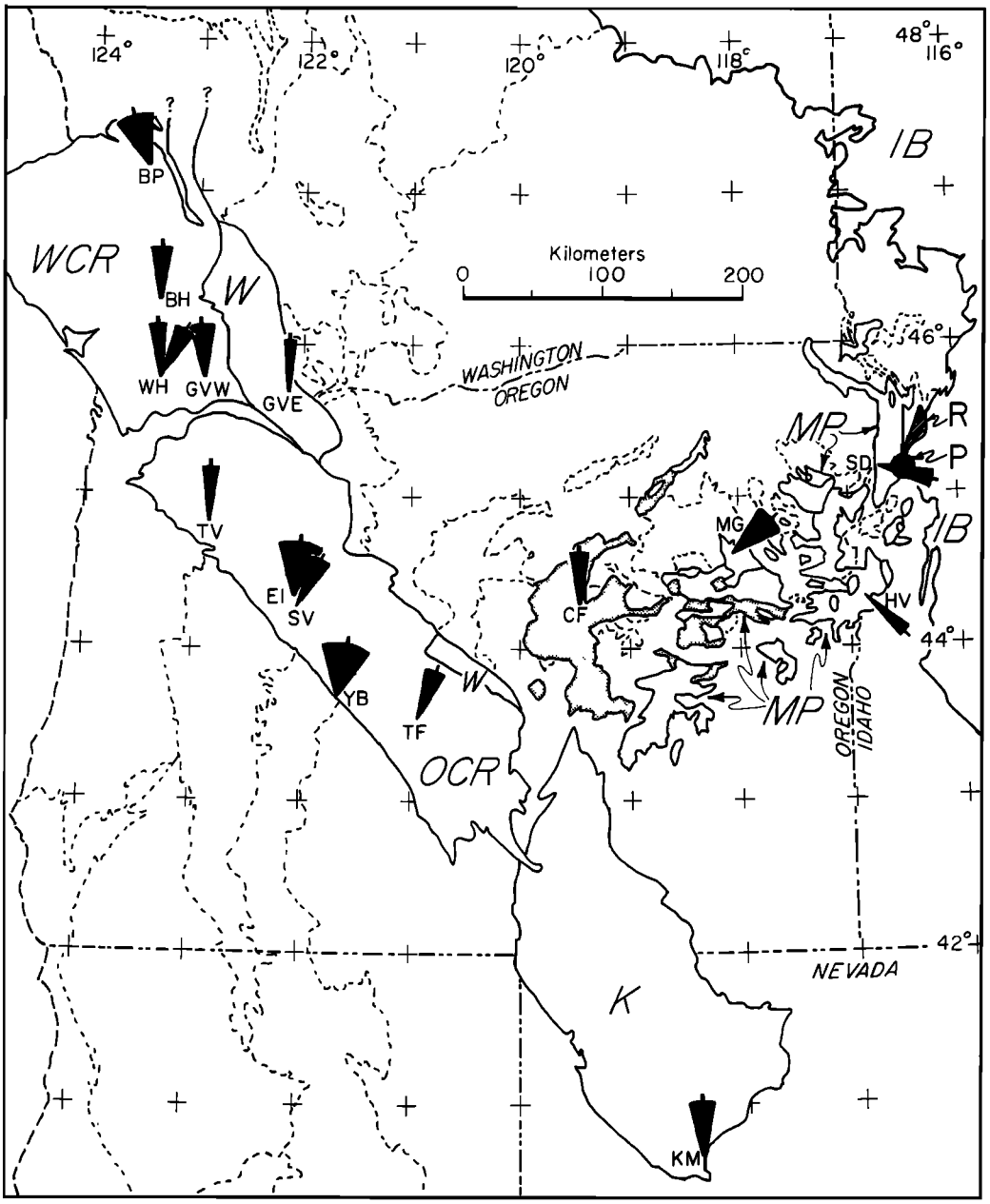


Fig. 6. Schematic reconstruction of part of the Pacific Northwest at approximately 38 m.y. Symbols for paleomagnetic data sites are as in Figure 1 and Table 1; data from rocks younger than the Clarno Formation are omitted. WCR, Washington Coast Ranges; OCR, Oregon Coast Ranges; W, parts of Western Cascades older than 38 m.y.; K, Klamath Mountains; CF, Clarno Formation (part); MP, Paleozoic and Mesozoic allochthonous terranes and intrusive rocks; IB, Idaho batholith and other cratonic rocks; R, Riggins Group. P is pivot point for rotation of pre-Tertiary basement of Blue Mountains province.

To the north and south the eastern contact of the Seven Devils Group is covered by the younger Columbia River Basalt Group. *Brooks and Vallier* [1978] have correlated the schist and amphibolite units of the Riggins Group with Jurassic and Permian-Triassic rocks in the Blue Mountains uplift. Following *Hamilton* [1976], Brooks and Vallier thus consider the Riggins Group to be partly or wholly accreted, and hence the pre-Mesozoic continental margin was in the vicinity of 116.2° west and is now largely obscured by intrusive rocks. Evidence for post-Miocene extensional tectonics in this part of western Idaho has been described by *Hamilton* [1962]. Between the Salmon River Mountains on the east and the Columbia Plateau on the west is a north-south trending zone of west-tilted blocks, bounded on their eastern sides by normal faults and mostly capped by the Columbia River Basalt Group. The total extension clearly is not great in terms of the scale of Figure 6. The close geologic ties among the Seven Devils Group, the Riggins Group, and the Idaho batholith, as are described above, require the east end of the Blue Mountains province to be near the batholith during Tertiary time. Thus we locate the pivot point for the tectonic rotation of the Blue Mountains province in the vicinity of the southern terminus of the exposed Riggins Group.

The reconstruction shown in Figure 6 was done as follows. The Blue Mountains province of Paleozoic and Mesozoic allochthonous terranes (MP), with its included Jurassic and Cretaceous plutonic rocks and its overlying remnants of the Clarno Formation, was rotated counterclockwise 16°, leaving

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51° (66.4° − 15.7°) of rotation in the plutonic rocks [*Wilson and Cox*, 1980] for pre-Clarno time. The main area of the Clarno Formation in central Oregon (CF) was also rotated counterclockwise 16° and was moved eastward toward the Paleozoic and Mesozoic province as far as possible, contracting the east-west extent of the Clarno by about 50 km. The Oregon Coast Ridge province including a pre-Oligocene fragment of the Western Cascades was rotated counterclockwise by 45°, satisfying the Tillamook Volcanics data (TV), and was then moved eastward close to the Clarno outcrop area. This unrotation also satisfies the data from the Eocene intrusive rocks (EI) and the Yachats Basalt (YB) but leaves 22° of rotation in the sedimentary rocks of the Tyee and Flournoy formations (TF) unremoved. The Washington Coast Range province, including a second fragment of the Western Cascades older than Oligocene, was rotated counterclockwise 28°, using the rotation indicated by the Goble Volcanics in the Western Cascades (GVE), and was joined as closely as possible to the north end of the Oregon Coast Range province. The data from the Goble Volcanics near the Willapa Hills (GVW) are also satisfied by this, and so are the data from two out of four of the Crescent Formation areas in this province, the Black Hills (BH) and one of the Willapa Hills (WH) subareas. Finally, the Klamath Mountains province was rotated 12° counterclockwise in accord with the rotation indicated by the Great Valley sequence data (KM) and was joined as well as possible to the southern end of the Oregon Coast Range province. Part, perhaps all, of the coastal Mesozoic province belongs to the Klamath Mountains in this reconstruction because it is overlapped by the Tyee basin rocks, but we have not shown it in Figure 6 because, as will be explained, it is difficult to draw a plausible picture of the original relationships of these three provinces with the data at hand.

Figure 6 is drawn as a rigid-block reconstruction, but this is only a convenience for the cartoonist. In fact, only two of the blocks were probably not internally deformed during their rotation and westward migration. The main outcrop area of the Clarno Formation (CF) is almost certainly a single structural unit. The part of the Oregon Coast Range province that contains the Tyee basin, i.e., from the southern border of the Tillamook Highlands south to the Klamath Mountains, may well have been a quasi-rigid block because of the consistency of current direction indicators and paleomagnetic declinations in the sedimentary rocks [*Snavely et al.*, 1964; *Simpson*, 1977; *Simpson and Cox*, 1977; *Heller and Ryberg*, 1983]. All that can be said about the Paleozoic and Mesozoic province exposed in the Blue Mountains uplift and eastward is that the magnetization direction in the Cretaceous-Jurassic plutons [*Wilson and Cox*, 1980] and the corresponding magnetic overprint in the Seven Devils Group [*Hillhouse et al.*, 1982] are uniform in direction over a large area, about 225 km in east-west extent.

The remaining two provinces (WCR and K) were certainly not rigid. *Frei et al.* [1984] have called attention to the gap that occurs between the Klamath Mountains and Coast Range provinces as a result of rigid-block reconstructions; in Figure 6 this gap is larger than in their 30-m.y. reconstruction. One way to eliminate the gap would be to suppose that the arcuate structural trends in the Klamath Mountains were straighter at 38 m.y. The only paleomagnetic data presently available [*Mankinen and Irwin*, 1982] come from the southern margin of the province (KM), and this hypothesis begs a test. Another way to account for the gap is to follow the suggestion of *Frei et al.* [1984] that deformation of the part of the coastal Mesozoic province which presently occupies most of the gap (Figure 1) accommodated all the relative rotations. As for the gap between the Washington and Oregon Coast Range provinces, this can be accounted for by internal deformation of the Washington part in the general area of the Willapa Hills (WH and GVW) [*Wells and Coe*, 1985]. The northern part of the Washington Coast Range province, i.e., the Olympic Mountains, is known to have been severely deformed in later Tertiary time, and this deformation is reflected in the paleomagnetic data on its eastern margin (BP) [*Beck and Engebretson*, 1982]. Hence the outlines of the Washington Coast Range and Klamath Mountains provinces as are drawn in Figure 6 must bear little resemblance to what they actually were at 38 m.y.

The reconstruction in Figure 6 is similar to several that have been previously published; it derives from the second model of *Simpson and Cox* [1977] with much more detail provided and with the room problem presented by the pre-Eocene rocks in eastern Oregon explicitly accounted for. The 30-m.y. reconstruction by *Frei et al.* [1984] is also quite similar, but we have split the Coast Range at the latitude of the Columbia River, and we do not deal with any structural provinces south of the Klamath Mountains. Our reconstruction differs from the one done by *Wells* [1984] in that we have closed the main outcrop area of the Clarno Formation against its eastern outlying remnants in the Blue Mountains uplift, greatly shrinking what is the present-day John Day Basin. This arrangement allows us to remove all of the post-Tyee rotation in both Coast Range provinces by the extension mechanism, whereas in the 37-m.y. reconstruction of *Wells* [1984], 23° was accounted for by simple shear.

The reconstruction proposed in Figure 6 implies that the Clarno Formation has been rifted apart and that volcanic rocks whose eruption should have resulted from such rifting would occur in the John Day Basin. The only candidates are the trachyandesite and basalt flows, and perhaps some of the ignimbrite sheets, that compose part of the John Day Formation [*Swanson and Robinson*, 1968]. The Monument dike swarm, which was the feeder complex for the Picture Gorge Basalt member of the Columbia River Basalt Group [*Swanson et al.*, 1979], also occurs mainly in the John Day Basin; perhaps these dikes represent the last stage of the proposed rifting. The plausibility of this aspect of our reconstruction could be judged by the reader by consulting, for example, the geologic map of *Walker* [1977].

Two more of the continuum of possible reconstructions will be briefly described; each is intended to address a particular aspect or problem associated with the one shown in Figure 6. The first alternative is designed to achieve the maximum eastward relocation of all the provinces. The same pivot point as in Figure 6 is used, but the Paleozoic and Mesozoic province is crowded as far as possible toward the Idaho batholith, thus taking out all but 29° of the rotation of the Jurassic and Cretaceous plutons. We would then close up the Clarno outcrop area to the same degree as before and would translate the western Tertiary provinces even farther east. Whereas the reconstruction in Figure 6 implies 300 km of expansion of the Basin and Range Province at latitude 42° since 38 m.y., this first alternate one implies 400 km of expansion. Either amount may very likely be more than is required by the geology of the Basin and Range Province itself.

A second alternative reconstruction would be to rotate the Paleozoic and Mesozoic allochthonous province 37° counterclockwise around the same pivot point as was done in the

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previous examples but not to reduce the original extent of the Clarno Formation as much as is done in Figure 6. Both of these (unillustrated) alternatives to Figure 6 would predict a difference in paleomagnetic declination of 21° between the main Clarno outcrop area, which we sampled, and the remnants of the Clarno Formation in the central Blue Mountains province, which we did not sample. The choice of reconstructions depends mainly on deciding how to distribute the post-Eocene extension, whether mainly in the John Day Basin, southeast of the Blue Mountains province, or equally in both areas.

CONCLUSIONS

A result of the reconstruction shown in Figure 6, and indeed of all reconstructions based on the concept of crustal extension as the cause of the observed rotations, is that the Cretaceous rocks in central Oregon that underlie the Clarno Formation (the Gable Creek and Hudspeth formations) are brought much closer to the partly correlative Hornbrook Formation which onlaps the northeastern flank of the Klamath Mountains, as is discussed by *Wells* [1984]. This stratigraphic similarity is the main potential geologic tie between the Blue Mountains province on the east and the Klamath Mountains and Coast Range provinces on the west; the only other control over relative paleolatitudes in our reconstruction comes from the paleomagnetic data themselves.

Another point concerns the possible significance of the existence of the Clarno Formation itself. According to *Heller and Ryberg* [1983], deposition of the Tyee Formation ceased at about 46 m.y., and the change within the Tyee basin from arkosic petrofacies to volcanolithic petrofacies occurred somewhat later, at about 45 m.y. Although the oldest reported date from the Clarno Formation is about 54 m.y., as is discussed above, the distribution of available ages implies that voluminous Clarno eruptive activity commenced at about 46 m.y. Perhaps the initiation of eruptive activity represented by the Clarno Formation may mark the time at which crustal extension was also initiated, and the extension that occurred while the Clarno was being deposited caused the first 22° of rotation of the existing Tyee basin. (The amount of rotation of the Tyee Formation (TF) not removed in the 38-m.y. reconstruction shown in Figure 6 is 22°.)

The paleomagnetic record of large-scale displacements in the eastern Columbia Embayment is now nearly enough complete that it can be used to characterize the entire tectonic history of the area. Anomalous inclinations and declinations in Triassic rocks of the Seven Devils Group and Huntington Formation show that they are an accreted fragment of the Wrangellia terrane [*Hillhouse et al.*, 1982] and that the Late Triassic and Early Jurassic continental margin was in the region now occupied by the Idaho batholith [*Brooks and Vallier*, 1978]. The paleomagnetic data from the Late Jurassic and Early Cretaceous plutons that intrude the allochthonous province show no paleolatitude anomaly, so that the rotation implied by their declination anomaly would have occurred during the last stages of accretion or during subsequent rifting [*Wilson and Cox*, 1980]. According to our reconstruction, all but about 16° to 37° of the rotations observed in the Jurassic and Cretaceous plutons, the Seven Devils Group, and the Huntington Formation are evidently the result of the last stages of accretion of these Paleozoic and Mesozoic terranes. In other words, the plutonism in the Blue Mountains province took place while that crustal block was still being transported eastward toward the craton. The main evidence for this interpretation is the Idaho batholith provenance of the sediments that filled the Tyee basin, as is discussed above [*Heller et al.*, 1985]. The discordant northeasterly trend of the Blue Mountains province (the "Idaho Orocline" of *Carey* [1958]) mostly dates from pre-Eocene time rather than being entirely the result of late Cenozoic extension of the Basin and Range Province, as was initially suggested by *Hamilton and Myers* [1966] and *Hamilton* [1969]. Only a small part of this discordant trend originated in post-Eocene time, and no large-scale rotations occurred in the eastern Columbia Embayment and the Blue Mountains province after the beginning of the eruption of the Columbia River Basalt Group in early Miocene time.

Virtually all post-Cretaceous and pre-Miocene paleomagnetic rotations in the PNW, at least at the latitude of present-day Oregon, can be accounted for by a fundamentally extensional model by assuming that extension also occurred within the Clarno Formation itself. The spatially progressive rotations recorded in various members of the western part of the Columbia River Basalt Group [*Magill et al.*, 1982; *Reidel et al.*, 1984; *Sheriff*, 1984] are better explained as the result of distributed shear [*Wells*, 1984]. Constraints on the extensional model that still need to be better evaluated are the geologic ties between the Washington Coast Ranges and provinces to the northeast, and the maximum plausible amount of extension of the northern Basin and Range Province.

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